

BIOMOLAI: A WEB-BASED PRECISION BIOINFORMATICS PLATFORM

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report “**BIOMOLAI: A WEB-BASED PRECISION BIOINFORMATICS PLATFORM**” is the bonafide work of “**Vipin Sudhakar, Rithvik Arulprakash, and Harshith KV**” who carried out the project work under my supervision.

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ABSTRACT

BioMolAI is a comprehensive, web-based bioinformatics suite developed to bridge the significant gap between raw biological sequence data and intuitive visual structural analysis. Traditionally, bioinformatics tools have imposed a steep learning curve, often requiring complex local software installations, command-line proficiency, or expensive computational hardware. This creates a barrier to entry for students and early-career researchers who need quick, accurate insights without the technical overhead.

This project solves that challenge by offering a lightweight, browser-based solution built on a modern stack of HTML5, Tailwind CSS, and JavaScript. The platform features a dual-module interface: a Protein Module and a Genomics Module. The Protein Module leverages the power of the ESMFold API (Meta AI) to predict and render 3D molecular structures in real-time using WebGL technology. Concurrently, it performs local algorithmic calculations for critical physicochemical properties such as molecular weight and isoelectric points. The Genomics Module simulates the Central Dogma of Molecular Biology, enabling users to translate DNA into protein while analyzing stability metrics like Melting Temperature. The result is a seamless, accessible "digital lab" that runs entirely in the browser.

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LIST OF SYMBOLS

SYMBOL	EXPANSION
API	Application Programming Interface
SPA	Single Page Application
PDB	Protein Data Bank
DNA	Deoxyribonucleic Acid
T_m	Melting Temperature
UI/UX	User Interface / User Experience

CHAPTER 1

INTRODUCTION

1.1 GENERAL OVERVIEW

Bioinformatics is an interdisciplinary field that develops methods and software tools for understanding biological data. As biological data accumulates, particularly in the form of DNA and protein sequences, the need for accessible visualization tools grows exponentially. Understanding the three-dimensional structure of a protein is crucial because structure dictates function; however, determining this structure experimentally via X-ray crystallography or NMR spectroscopy is time-consuming and expensive.

BioMolAI is designed as a browser-based "Studio Edition" tool that allows users to analyze biological sequences and visualize them in 3D without the need for heavy software installations like PyMOL or Chimera. By leveraging modern web technologies, the computational load is handled efficiently in the browser and via cloud APIs, democratizing access to structural biology.

1.2 OBJECTIVES

The primary objectives driving this project are:

- **3D Visualization:** To enable the rendering of 3D protein structures directly from amino acid sequences using cloud-based inference (ESMFold) and WebGL technology.
- **Physicochemical Analysis:** To automate the calculation of key protein properties, including Isoelectric point (pI), Molecular Weight, and Extinction Coefficients, which are vital for wet-lab experiment planning.
- **Genomic Simulation:** To digitally simulate the Central Dogma of Molecular Biology (Replication → Transcription → Translation) for educational purposes.
- **Modern UI/UX:** To provide a high-fidelity, aesthetically pleasing user interface (UI) that enhances user engagement.

1.3 SCOPE OF THE PROJECT

The scope of BioMolAI is focused on the analysis of single-chain proteins and standard DNA sequences. It is designed primarily for educational purposes and preliminary research analysis. The system utilizes the ESMFold API for structure prediction, necessitating an active internet connection. We focused on integrating existing high-performance APIs into a seamless user experience rather than developing novel folding algorithms from scratch.

CHAPTER 2

LITERATURE REVIEW

2.1 THE PROTEIN FOLDING PROBLEM

The "Protein Folding Problem"—determining how a protein's amino acid sequence dictates its unique 3D atomic structure—has challenged biologists for over 50 years. Recently, the field has been revolutionized by the application of Artificial Intelligence.

AlphaFold2 (DeepMind): In 2020, AlphaFold2 achieved a breakthrough, solving the folding problem with atomic-level accuracy. However, running AlphaFold2 locally requires significant GPU resources and technical expertise, making it difficult for students and undergraduate labs to utilize effectively.

ESMFold (Meta AI): More recently, Large Language Models (LLMs) trained on protein sequences have emerged as a viable alternative. ESMFold offers a distinct speed advantage, predicting structures up to 60x faster than AlphaFold2 while maintaining comparable accuracy for many targets. We selected ESMFold for this project due to its low latency and API accessibility, which makes it uniquely suitable for real-time web applications.

2.2 EXISTING WEB TOOLS

While many tools exist, the current landscape of web-based bioinformatics often suffers from outdated User Interfaces and fragmented functionality.

- **RCSB PDB:** The Protein Data Bank is the gold standard for storing structures, but it primarily serves as a repository for experimentally determined data. It does not typically predict structures from raw text sequences on the fly.
- **ExPASy ProtParam:** A widely used tool for calculating physicochemical properties. While highly accurate, it is purely text-based, lacks visualization capabilities, and features an interface that has not been updated in decades.

BioMolAI integrates the calculation capabilities of tools like ProtParam with the advanced visualization power of modern WebGL libraries (3Dmol.js) into a single, cohesive Single Page Application (SPA).

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 SYSTEM ARCHITECTURE

To ensure a seamless user experience, the application follows a modern **Single Page Application (SPA)** architecture. This approach ensures high responsiveness, allowing users to switch between modules without the friction of page reloads.

The system architecture is divided into three distinct layers:

1. **Presentation Layer:** Built with HTML5 and Tailwind CSS, this layer handles the visual structure, responsive layout, and theming.
2. **Logic Layer:** Vanilla JavaScript (ES6) serves as the engine, handling user input validation, performing local calculations (Molecular Weight, pI, T_m), and managing the application state.
3. **Service Layer:** The external ESMAtlas API processes the computationally intensive task of protein folding, returning PDB data which is then rendered by the browser.

3.2 MODULE DESCRIPTION

3.2.1 Protein Analysis Module

This module functions as the core of the application. It accepts inputs in FASTA format or via predefined samples (e.g., Insulin, PETase). The system performs statistical analysis by iterating through the amino acid chain. It calculates molecular weight by summing atomic masses defined in a constant dictionary and estimates the isoelectric point (pI) based on the ratio of positive (Arg, Lys, His) to negative (Asp, Glu) residues.

3.2.2 Genomics and Central Dogma Module

This module is dedicated to DNA sequence analysis. It calculates the Melting Temperature (T_m) using standard formulas, which is critical for PCR applications. Furthermore, it simulates the biological processes of transcription (replacing Thymine with Uracil) and translation (mapping codons to amino acids), visualizing the flow of genetic information.

3.3 UI/UX DESIGN STRATEGY

Scientific tools often neglect user experience, leading to clunky workflows. BioMolAI prioritizes "scannability" and modern aesthetics to reduce cognitive load for the user.

Theme Engine: A built-in toggle allows users to switch between "Light Mode" (Clean/Clinical) and "Dark Mode" (Deep Void). This logic is implemented using CSS variables and the Tailwind 'dark:' modifier, ensuring comfort during both day and night usage.

Reactive Background: To create an immersive "lab" atmosphere, the application features an HTML5 Canvas background. A grid of dots renders in the background, and JavaScript event listeners track the mouse position. The Euclidean distance between the cursor and each dot is calculated in real-time, creating a subtle, interactive biological effect.

3.3.1 Input Constraints

To ensure system stability and data validity, the following constraints are strictly enforced:

Data Type	Constraint	Reasoning
Protein Sequence	Min 10, Max 400 residues	Prevents API timeouts and visualization lag.
DNA Sequence	A, T, C, G characters only	Ensures biological validity of the input.
API Requests	Debounced / Button Lock	Prevents excessive calls to the free ESMFold tier.

Table 3.1: Input constraints for Biological Sequences

CHAPTER 4

IMPLEMENTATION

4.1 TECHNOLOGY STACK

We selected a robust technology stack designed for performance and ease of deployment:

- **HTML5:** Provides the semantic structure and accessibility foundation.
- **Tailwind CSS:** A utility-first CSS framework chosen for its ability to rapidly build responsive, glassmorphism-style interfaces without writing custom CSS files.
- **JavaScript (ES6+):** Handles all client-side logic, including asynchronous `fetch` requests and dynamic DOM manipulation, without the overhead of heavy frameworks like React.
- **3Dmol.js:** A specialized WebGL-based library designed specifically for high-performance rendering of molecular data (PDB files) within standard web browsers.

4.2 ALGORITHMIC LOGIC

4.2.1 Physicochemical Calculations

The analysis logic is encapsulated in the ``analyzeProteinStats()`` function. It utilizes a predefined dictionary of Amino Acid atomic masses. The algorithm iterates through the input string with $O(n)$ time complexity, summing masses and classifying residues as hydrophobic, charged, or polar to generate the composition charts.

4.2.2 3D Rendering Integration

The 3D rendering workflow is handled by the ``predictStructure()`` function:

1. **Sanitization:** The user input is cleaned of whitespace and non-standard characters.
2. **Request:** An asynchronous POST request is sent to ``https://api.esmatlas.com/foldSequence/v1/pdb/``.
3. **Parsing:** The API returns a PDB (Protein Data Bank) string, which describes the 3D coordinates of every atom.
4. **Rendering:** The ``3Dmol.js`` viewer parses this string and applies a "cartoon" visualization style. Residues are colored based on the **b-factor** (pLDDT confidence score), providing immediate visual feedback on prediction quality.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 PROTEIN RENDERING RESULTS

The application was rigorously tested using various biological sequences. Complex proteins, such as the Green Fluorescent Protein (GFP), were successfully visualized in 3D. The rendering engine accurately interpreted secondary structures, clearly displaying Alpha helices and Beta sheets.

The visualization latency was remarkably low (approximately 1-2 seconds for folding via API), validating our choice of ESMFold over a local AlphaFold installation for educational use cases. The confidence-based color coding (Blue for high confidence, Red for low) provided users with immediate, intuitive feedback regarding the reliability of the generated structure.

5.2 GENOMICS ANALYSIS RESULTS

The Genomics Module accurately simulated the central dogma process. During testing with the Human Insulin gene sequence, the system correctly identified the Start Codon (ATG) and the Stop Codon (TAG), ensuring accurate translation.

The calculated Melting Temperature (T_m) provides users with critical data for planning PCR experiments. The formula used, $T_m = 64.9 + 41 * (GC - 16.4) / Length$, yielded results consistent with standard industry calculators. Furthermore, the integration between the DNA and Protein modules functioned seamlessly; users could translate a DNA sequence and immediately fold the resulting protein, effectively demonstrating the biological "Replication to Structure" pipeline.

CHAPTER 6

CONCLUSION

6.1 CONCLUSION

BioMolAI successfully demonstrates how modern web technologies can democratize access to advanced scientific tools. By combining a refined, intuitive user interface with powerful algorithmic logic, the project serves as a robust educational and analytical platform for bioinformatics. It completely eliminates the need for complex software installations, making structural biology accessible to any student with a web browser. The project met all primary objectives: 3D visualization, physicochemical analysis, and central dogma simulation were all implemented and verified successfully.

6.2 FUTURE SCOPE

While the current version is fully functional, we have identified several key enhancements for future iterations:

- **File Export:** Adding functionality to download the generated `.pdb` files, allowing users to take their data into other professional software.
- **Multiple Chains:** Extending support to predict protein complexes (quaternary structure) rather than just single chains.
- **Mutation Analysis:** Developing a feature to highlight how single base-pair changes (SNPs) affect the final protein structure, which is crucial for medical research.

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APPENDICES

APPENDIX 1: SOURCE CODE SNIPPET (DNA ANALYSIS)

The following JavaScript function demonstrates the logic used to calculate GC Content and Melting Temperature from a raw DNA string.

```
// Function to analyze DNA Stats
function analyzedDNA() {
  // 1. Get Input and Sanitize (Remove spaces, uppercase)
  const raw = document.getElementById('dna-input').value
    .toUpperCase().replace(/\s/g, '');

  // 2. Validate Input (Regex for ATCG only)
  if(! /^[ATCG]+$/.test(raw)) return alert("Invalid DNA");

  // 3. Calculate GC Content
  const gc = (raw.match(/[CG]/g) || []).length;

  // 4. Calculate Melting Temp (Standard Formula)
  const tm = (64.9 + 41*(gc - 16.4)/raw.length).toFixed(1);

  // 5. Update DOM
  document.getElementById('stat-gc').textContent =
    (gc/raw.length*100).toFixed(1)+'%';
  document.getElementById('stat-tm').textContent = tm+'°C';
}
```